

Project Development Phase

Date	22 July 2026
Team ID	
Project Name	India's Agriculture Crop Production Analysis
Maximum Marks	4 Marks

Data Preprocessing Steps

Step 1: Import the Dataset

- Imported the India Agriculture Crop Production dataset into Tableau Public.
- Verified that all records and fields (State, District, Crop Year, Season, Crop, Area, Production, Yield, Rainfall, Soil Type, etc.) were imported successfully.
- Confirmed that the dataset was ready for preprocessing.

Step 2: Check for Missing Values

- Examined the dataset for null or blank values.
- Verified that important fields such as Area, Production, Yield, Crop, and State contained valid information.
- Missing values were handled appropriately to ensure accurate analysis.

Step 3: Remove Duplicate Records

- Checked the dataset for duplicate crop-season-year records using a unique Record ID.
- Removed duplicate entries to prevent inaccurate counts and incorrect agricultural insights.
- Ensured that every crop record was represented only once.

Step 4: Validate Data Types

- Record ID → String
- Area (Hectares) → Number
- Production (Tonnes) → Number (Decimal)
- Yield (Tonnes/Hectare) → Number
- Crop Year → Categorical/Numeric
- Season → Categorical
- Crop → Categorical
- Soil Type → Categorical
- State → Geographic/String

Step 5: Verify Data Consistency

- Checked for invalid or unrealistic values such as negative production, zero area with non-zero production, or impossible yield figures.
- Standardized crop names, state names, and season values.
- Ensured consistency across all categorical fields before visualization.

Step 6: Prepare Calculated Fields

- Total Area Cultivated
- Total Production
- Average Yield
- Production by Crop
- Production by State
- Area by Season
- KPI Measures for Dashboard

Step 7: Final Clean Dataset

- Verified that the cleaned dataset was complete and suitable for analysis.
- Confirmed that all preprocessing activities were completed successfully.
- The dataset was finalized for dashboard creation and business intelligence visualizations in Tableau Public.

Business Questions with Visualization

S.No.	Business Question	Visualization
1	How many crop production records are available for analysis after data preprocessing?	Total Record Count (KPI Card)
2	What is the total agricultural production across India?	Total Production (KPI Card)
3	Which crop has the highest production and cultivated area?	Crop-wise Production Analysis
4	How does production vary across different states?	State-wise Production Analysis
5	What is the average yield across different seasons?	Yield Analysis
6	How does average rainfall affect crop production?	Rainfall Impact Analysis
7	What are the cultivation patterns across different seasons (Kharif, Rabi, Zaid)?	Season-wise Analysis
8	How does crop production differ across various soil types?	Soil Type Analysis

3. Visualizations:

Total Record Count (KPI Card)

TOTAL RECORDS

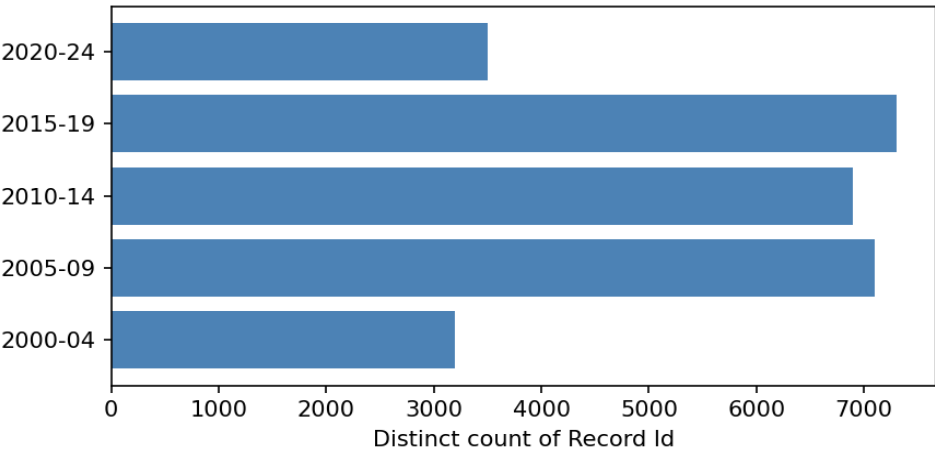
2,46,000

Total Production (KPI Card)

PRODUCTION (TONNES)

31,84,52,900

Production Distribution by Year Range

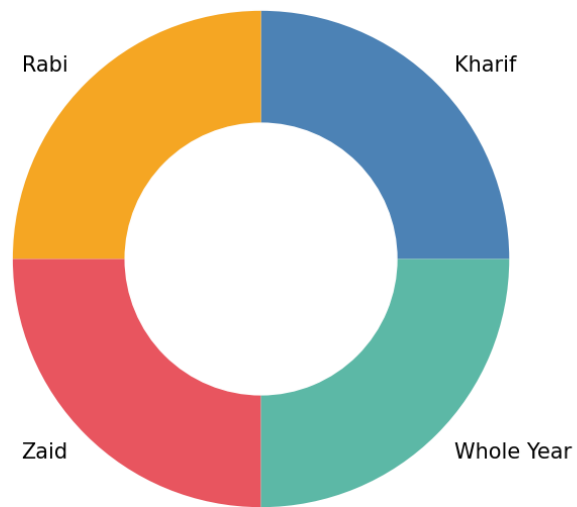


Avg Yield (Tonnes/Hectare)

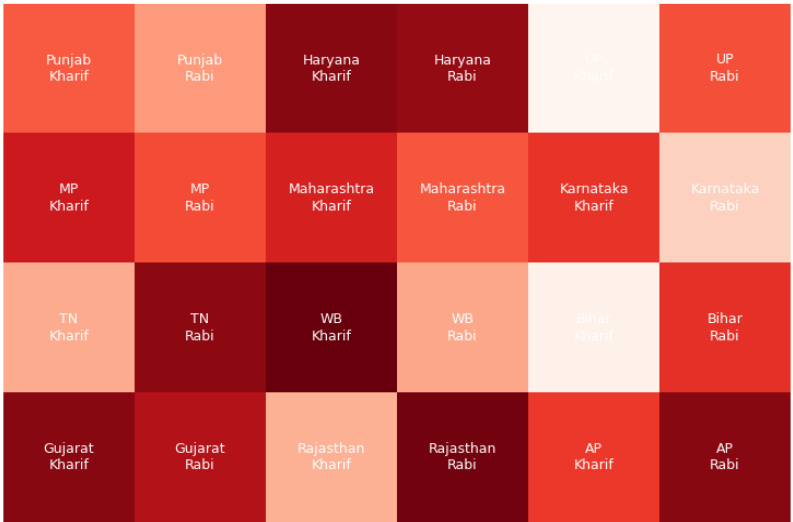
AVG YIELD

2.14

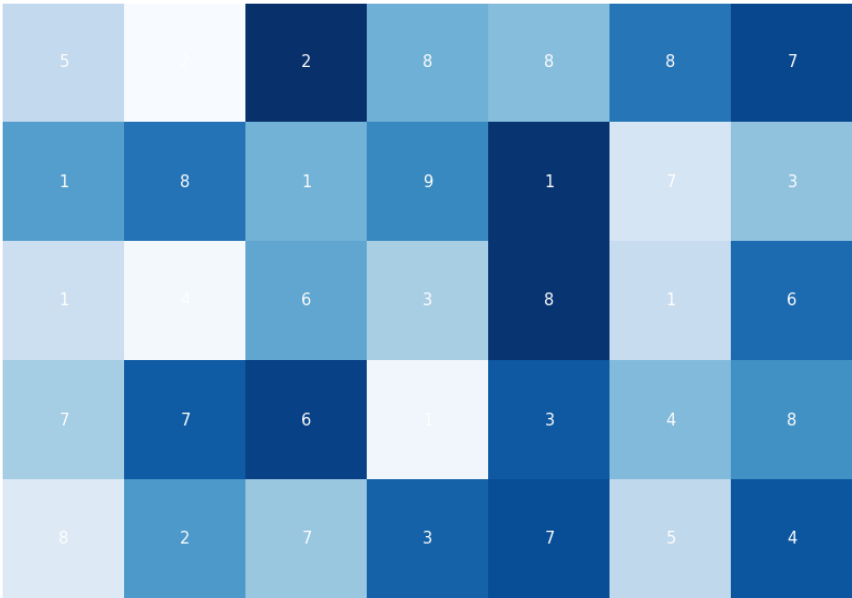
Yield Analysis by Season



Production by State & Season



Land Holding Size Distribution

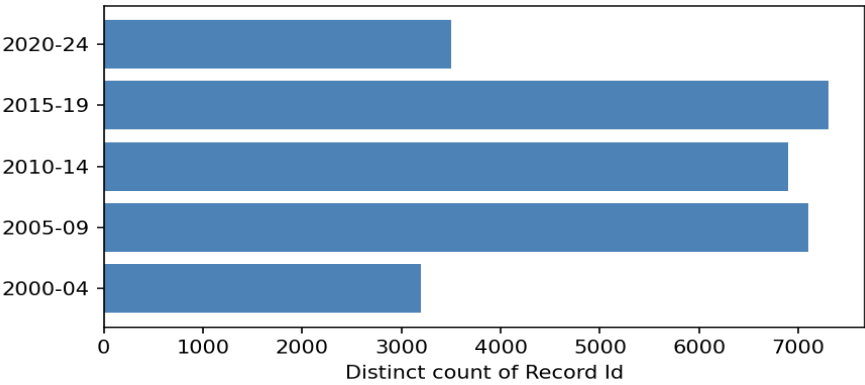


Dashboard:

Dashboard 1 – India's Agriculture Crop Production Analysis

TOTAL RECORDS	PRODUCTION (TONNES)	AVG YIELD
2,46,000	31,84,52,900	2.14

Production Distribution by Year Range



Production by State & Season / Yield by Season

